

Project Estimation Starter

Team 10 | CS 4390/5388 | Campus Dining

Portal record: September 29, 2026

Starter · Step 1: What Release 1 must do

T = Testing; S = Security or privacy; I = Integration with other systems; D = Data (collection, cleaning, migration); H = Hosting and deployment; U = Usability and accessibility; O = Documentation or training.

Release 1 capabilities and hidden work

#	Capability	Evidence	Hidden work
1	Students can view campus dining locations, hours, and whether each location is open.	Interview commuting sophomore	T, I, D, H, U
2	The system works on phones and provides information that is easy to scan.	Interview time pressed students	T, H, U
3	Students can find sold-out items within each dining location's menu.	Interview X,	T, I, D, H, U, O
4	Students can filter results for their dietary restrictions or preferences.	Vegetarian, allergy, and meal plan participants described these constraints.	T, D, U
5	Students can view a current crowd level and estimated wait time for each location.	Commuting sophomore chose the shortest line	T, I, D, H, U

The sold-out item capability still cites "Interview X" in the starter. The team needs to identify that interview.

The starter calls this list "Release 1." The current charter covers research, requirements, and prototype evaluation. This list describes possible system capabilities; it does not approve a production release.

Starter · Step 2: Our own experience

A similar project our team has built

Question	Our experience
What was it?	Streaming-App (Music/Video)
How long was it expected to take?	3 months
How long did it take?	3 months
What took longer than expected?	Coordinating GitHub changes, integrating team content, responsive layout testing, and deployment details.

Starter · Step 3: Similar products

Product A (small): Waitwhile

Product A (small): Waitwhile

Field	Our research
Product and company	Waitwhile - Waitwhile
Why it is similar to our project	Lets customers join a virtual line and wait elsewhere instead of standing in a crowded location.
Development started	Unknown. Waitwhile's company press page gives 2017 as the founding year; it does not establish when product development started.
First public release	2016, according to Waitwhile's April 19, 2020 product blog.
Time to first release	Unknown: Development start date not verified.
What the first release included	A virtual waitlist for managing customers' place in line
Team size, if known	Two co-founders are identified; total launch-team size unknown.
What worked	Remote queuing addressed the immediate problem of standing in line; the product later attracted many business customers.
What did not work	Its original platform later became slow and had outages at scale, prompting a rebuild.

Sources: Waitwhile product launch and rebuild; Waitwhile company history; Waitwhile LinkedIn profile; Equilar company profile.

Product B (big): Grubhub

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Field	Our research
Product and company	Grubhub - Grubhub
Why it is similar to our project	Lets customers find restaurants and order takeout or delivery instead of waiting at a busy restaurant. It does not primarily predict crowd levels.
Development started	2004
First public release	2004 in Chicago.
Time to first release	Within 2004; exact number of months unknown.
What the first release included	A website where users can enter an address to find restaurants that are delivered there and view delivery information and menus.
Team size, if known	Two founders, Matt Maloney and Mike Evans; total launch-team size unknown.
What worked	Address-based search answered a specific question: "Who delivers to me?"
What did not work	Ordering food avoids a restaurant visit but does not tell users how crowded a place is or help with stores.

Sources: Company history; Grubhub history.

Product C: Google Maps

Product C: Google Maps

Field	Our research
Product and company	Google Maps - Google/Alphabet
Why it is similar to our project	Helps users avoid crowded restaurants and stores using historical and live busyness data.
Development started	2003
First public release	February 8, 2005
Time to first release	Approximately 2 years, based on a 2003 start and February 2005 launch.
What the first release included	A desktop website with maps and directions. Popular Times was not part of the first release.
Team size, if known	Four people are identified with the original Where 2 Technologies team; Google's launch-team size is unknown.
What worked	Simple maps and directions gave users a useful starting point; later business information added a way to plan visits.
What did not work	Business information is unavailable when Google lacks sufficient visit data; it is not an exact headcount.

Sources: Google Maps launch history; Popular Times data and limitations; Google Maps history.

Waitwhile's first product launch and its company founding are separate dates in its official sources. Neither source establishes the development start date, so time to first release remains unknown.

Starter · Step 4: What we took from the comparison

Hidden work

T (Testing) and U (Usability and accessibility) each appear in all five rows. Most of the products we have been involved with required thorough testing of features or workflows.

Time and team size

Usually a rushed 3 months, with a group of 2 to 3 team members. This suggests that, while feasible within the time scale of this class, a first release would still require a large number of work hours to reach a first release candidate.

What we will copy and avoid

We will copy business information such as open hours and options like gluten-free so people can plan where they want to visit and not waste their time. A mistake we will avoid is thinking too small about the scale, which could lead to a rebuild and outages.

Our first rough estimate

Our comparables suggest a first release would take 2 to 3 months with a team of 4.

The 3-month experience and the 2-to-3-month release estimate are the team's starter answers. The comparable-product table does not establish a shared 3-month development timeline. Activity 2 estimates a smaller slice using function points and use-case points.